

(log)

PURE

LOGARITHMS AND EXPONENTIAL FUNCTIONS

LOGARITHMS (logs)

A log is a POWER of a BASE NUMBER.

Every power statement has an equivalent log statement.

$$\begin{array}{ccc} \begin{array}{c} \text{Power} \nearrow \\ 10^2 = 100 \\ \text{base} \nwarrow \end{array} & \Leftrightarrow & \begin{array}{c} \text{Power} \nearrow \\ \log_{10} 100 = 2 \\ \text{base} \nwarrow \end{array} \end{array} \quad \begin{array}{l} \text{log base 10} \\ \\ \end{array}$$

$$2^3 = 8 \quad \Leftrightarrow \quad \log_2 8 = 3 \quad \begin{array}{l} \text{log base 2} \\ \\ \end{array}$$

In general $b^a = x \quad \Leftrightarrow \quad \log_b x = a$

$$a^0 = 1$$

$$\log_a 1 = 0$$

Eg. solve for x

$$1) \log_7 2401 = x \quad (7^4 = 2401)$$

$$x = 4$$

$$2) \log_2 0.25 = x \quad (2^{-2} = 0.25)$$

$$x = -2$$

Log Rules

1. $\log(a \times b) = \log(a) + \log(b)$

e.g. $\log(300) = \log(3) + \log(100)$

$\uparrow$
 3×100

$2.477 = 0.477 + 2$

→ if no base is shown,
then it is true for any base.

2. $\log(a \div b) = \log(a) - \log(b)$

e.g. $\log(3) = \log(15) - \log(5)$

$\uparrow$
 $\log\left(\frac{15}{5}\right)$

3. $\log(a^n) = n \times \log(a)$

e.g. $\log(2^5) = 5 \times \log(2)$

$1.505 = 5 \times 0.3010$
 $= 1.505$

4. $\log_b(1) = 0$

$b^0 = 1$

5. $\log_b b = 1$

$b^1 = b$

$\log_{53} 53 = 1$

log

↑ If no base is shown,
Then it is true for any
base.

Sometimes

$\log(5) = 0.6989...$
mean

$\log_{10}(5) = 0.6989... \text{ (log base 10)}$

6. $\log_b x = \frac{\log_a x}{\log_a b}$

e.g. $\log_7 50 = \frac{\log_{10} 50}{\log_{10} 7}$

$$6. \log_b x = \frac{\log_a x}{\log_a b}$$

e.g

$$\log_7 50 = \frac{\log_{10} 50}{\log_{10} 7}$$

Using Logs to solve equations

If $b^x = n$, then $x = \log_b n$

$$2^x = 1000000$$

$$x = \log_2 1000000$$

$$\text{or } \frac{\log(1000000)}{\log(2)}$$

$$= 19.9 \text{ (3sf)}$$

NOTE: If $b^{ax+c} = n$ use this approach:

$$\text{eg } 2^{4x+3} = 1000$$

Take $\log_{10}$ of both sides

$$\log_{10}(2^{4x+3}) = \log_{10}(1000)$$

$$(4x+3) \times \log_{10}(2) = \log_{10}(1000)$$

$$4x+3 = \frac{\log_{10}(1000)}{\log_{10}(2)}$$

$$4x+3 = 9.96578...$$

$$x = \frac{9.96578... - 3}{4}$$

$$x = 1.74144...$$

$$= 1.74 \text{ (3sf)}$$

$$\log(5^3) = 3 \times \log(5)$$

$$\log_2 1000$$

Simplifying log expressions

This means writing with the least number of log terms

Eg

$$\text{Simplify } (\log 4 + \log 6) - \log 3 \quad \text{use log rules}$$

$$= \log(4 \times 6) - \log 3$$

$$= \log(24) - \log 3$$

$$= \log\left(\frac{24}{3}\right)$$

$$= \log(8)$$

$$\begin{aligned} \log 2^3 \\ 3 \log 2 \end{aligned}$$

use log rules backward

$$\log\left(\frac{24}{3}\right) = \log 24 - \log 3$$

$$\log(a \times b) = \log a + \log b$$

$$\begin{aligned} \log(a \times b) \\ \uparrow \\ = \log a + \log b \end{aligned}$$

$$3 \log 4 - 2 \log 2 = \log(4^3) - \log(2^2)$$

$$= \log\left(\frac{4^3}{2^2}\right)$$

$$= \log(16)$$

$$\frac{5}{2} \log 4 - \frac{1}{2} \log 64 = \log\left(4^{\frac{5}{2}}\right) - \log\left(64^{\frac{1}{2}}\right)$$

$$= \log 32 - \log(8)$$

$$= \log\left(\frac{32}{8}\right)$$

$$= \log 4$$

log inequalities

used when solving $b^x < a$

Eg solve $2^n > 1000$

take logs

$$\log(2^n) > \log(1000)$$

$$n \log(2) > \log(1000)$$

$$n > \frac{\log(1000)}{\log(2)}$$

If $n > 9.97$, it will > 1000

$$n > 9.97 \text{ (3sf)}$$

Solve $(0.92)^n < 0.1$

$$\log(0.92^n) < \log(0.1)$$

(-) ย้ายข้างเลข, แล้วหาค่า

$< 1 = (-)$

$$n \log(0.92) < \log(0.1)$$

negative

be careful

$$n > \frac{\log(0.1)}{\log(0.92)}$$

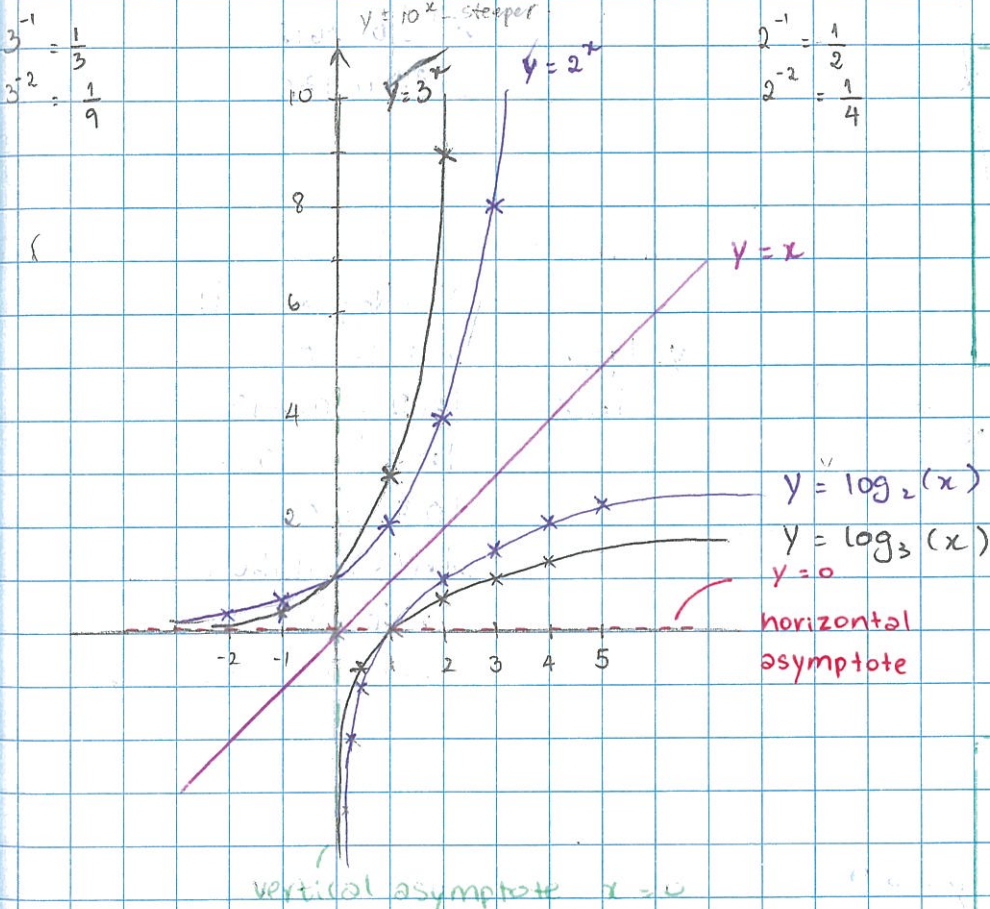
$$n > 27.6 \text{ (3sf)}$$

any amount > 27.6 , < 0.1

Exponentials and Logs

An exponential function is $y = a \times b^x$
where a and b are constants.

eg. $y = 2^x$ ($a = 1, b = 2$)



similarities (if $b > 1$)

- Shape
- horizontal asymptote $y = 0$
- y-intercept is $y = 1$

log graph similarities

- Shape
- vertical asymptote $x = 0$
- x-intercept is $x = 1$

$\log_2(1) = 0$

$\log_2(2) = 1$

$(3) = 1.58$

$(4) = 2$

$(5) = 2.32$

$\log_2(0.5) = -1$ (2^{-1})

$\log_2\left(\frac{1}{4}\right) = -2$

$(0.1) = -3.32$

$\log_3(1) = 0$

$(2) = 0.63$

$(3) = 1$

$(4) = 1.26$

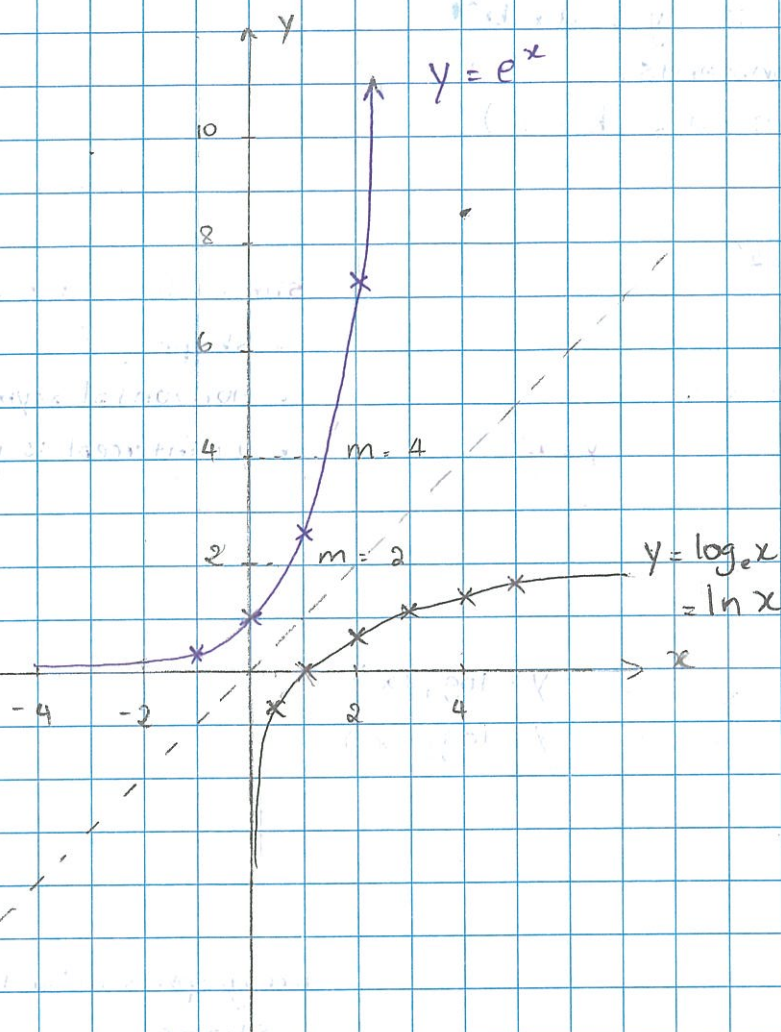
$(5) = 1.46$

$(0.5) = -0.63$

$(0.2) = -1.46$

NOTE: $y = b^x$ and $y = \log_b x$
are inverse functions
(reflections in line $y = x$)

The Graph of $y = e^x$ (where $e \approx 2.71828...$)



$$y = e^x$$

$$e^1 = 2.7182$$

$$e^2 = 7.389$$

$$e^3 = 20.0855$$

$$e^0 = 1$$

$$e^{-1} = 0.3678$$

$$e^{-2} = 0.135$$

$$y = \log_e x = \ln$$

$$\ln 1 = 0$$

$$\ln 2 = 0.693$$

$$\ln 3 = 1.098$$

$$\ln 4 = 1.386$$

$$\ln(5) = 1.609$$

$$\ln(0.5) = -0.693$$

$$\ln(0.25) = -1.386$$

Special property of $y = e^x$

- At any point on the curve, the gradient (m) is equal to the y -value.

The "natural log" $y = \ln x$

The term "natural log" is used instead of $\log_e$
ie $\log_e x = \ln x$

(n-natural)

So, $\ln 5 = 1.609...$ $\ln(e) = 1$
means $e^{1.609...} = 5$

Eg. Solve

$$5e^x + 4 = 100$$

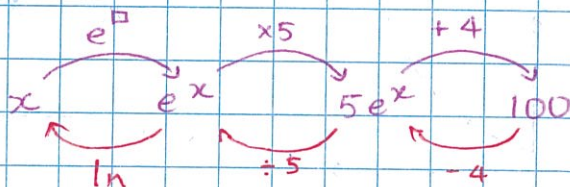
$$5e^x = 96$$

$$e^x = \frac{96}{5}$$

$$x = \ln\left(\frac{96}{5}\right)$$

$$x = 2.9549...$$

$$x = 2.95 \quad (3sf)$$



Eg. $3 \ln(x-2) = 50$

$$\ln(x-2) = \frac{50}{3}$$

$$x-2 = e^{\left(\frac{50}{3}\right)}$$

$$x = e^{\left(\frac{50}{3}\right)} + 2$$

$$x = 17307781.95...$$

$$x = 1.73 \times 10^7 \quad (3sf)$$

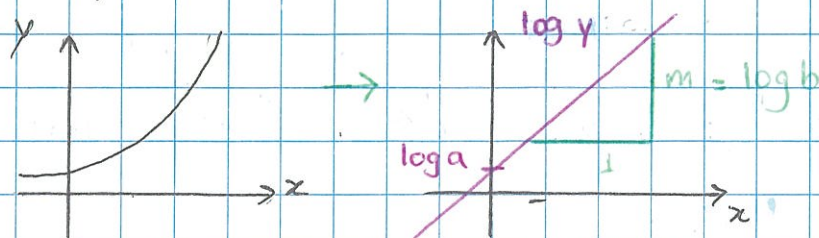
Log Modelling

(ie using logs to find the equations of exponential or power curves)

1. Exponential curves

$$y = ab^x \quad \text{where } a, b \text{ constants}$$

eg. $y = 2.69 \times 3.1^x$



if $y = ab^x$

then the graph of $\log y$ vs x will give a straight line
with vertical intercept = $\log a$
gradient = $\log b$

$\Rightarrow$ Proof: if $y = ab^x$

$$\text{then } \log(y) = \log(ab^x)$$

$$= \log(a) + \log(b^x)$$

$$= \log(a) + x \log(b)$$

$$\log(y) = \underbrace{[\log(b)]}_{m}x + \underbrace{[\log(a)]}_{c} \rightarrow \text{form of straight line}$$

$$y = 1 \times 3^x$$

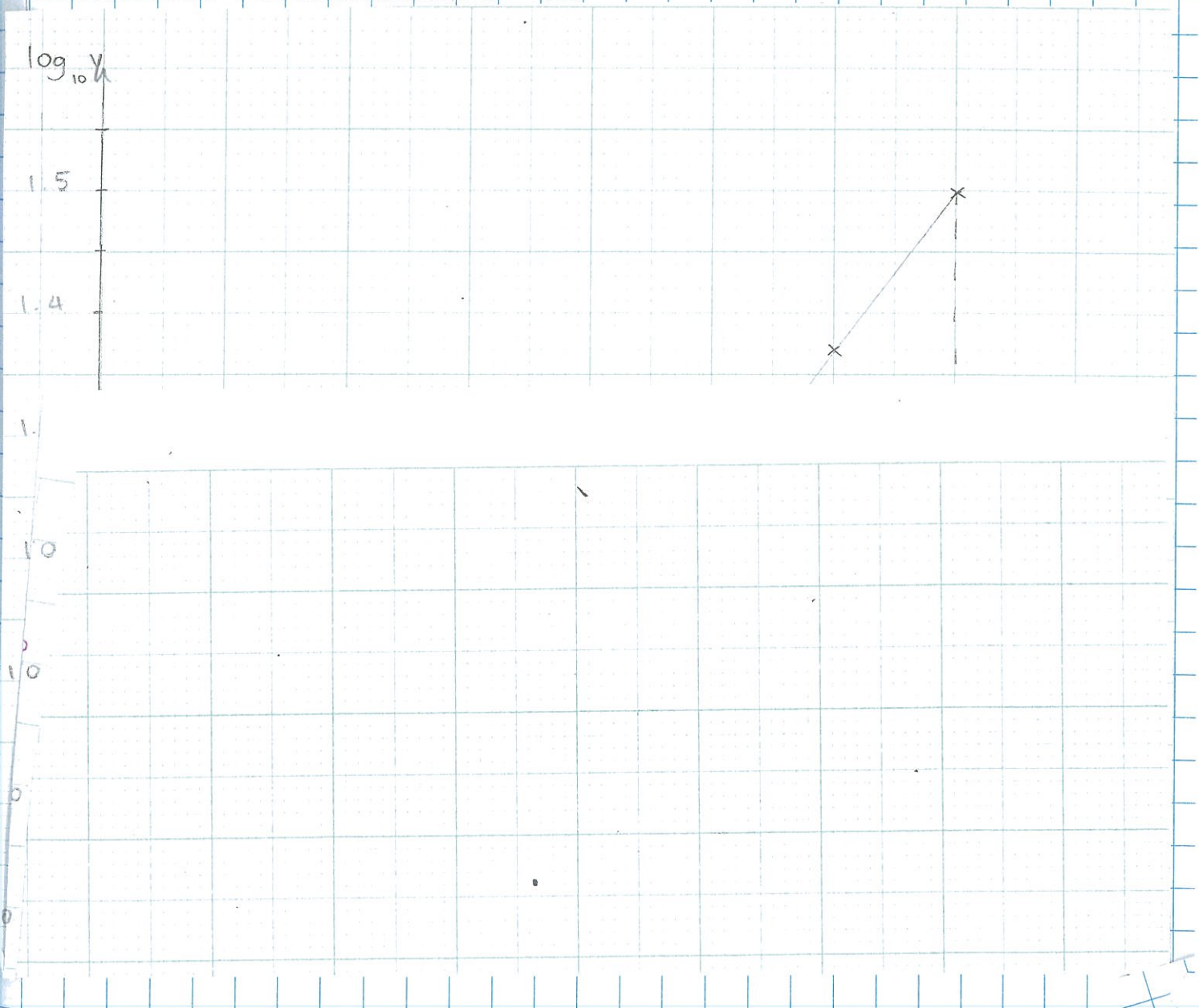
$$\log_{10} a = 0$$

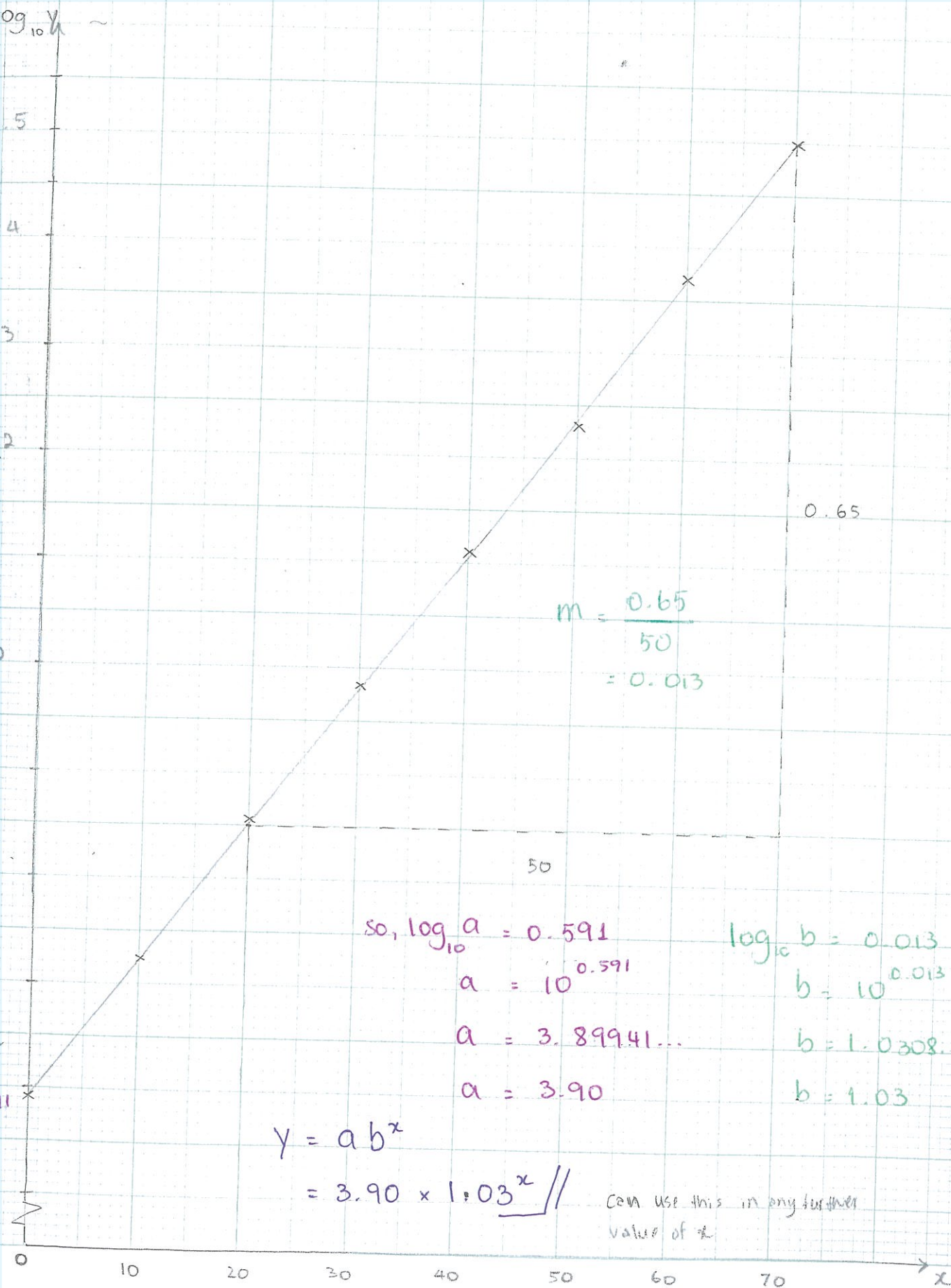
$$a = 10^0 = 1$$

Eg. Population of USA

Year	Time (yr) (x)	Pop. (millions) (y)	^{3.s.f} $\log_{10}(y)$
1790	0	3.9	0.591
1800	10	5.3	0.724
1810	20	7.2	0.857
1820	30	9.6	0.982
1830	40	12.9	1.11
1840	50	17	1.230
1850	60	23.2	1.37
1860	70	31.4	1.50

Find equation assuming $y = ab^x$



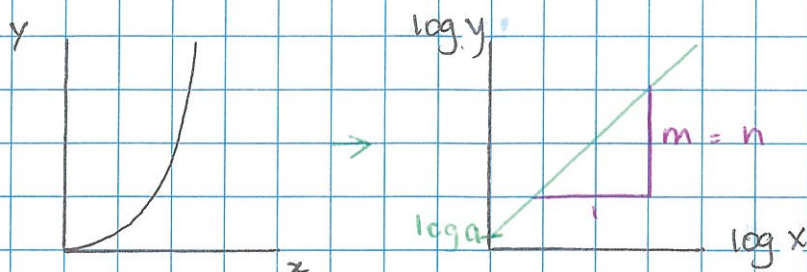


2. Power Curve

$$y = ax^n$$

a, n constants

e.g. $y = 3x^5$



if $y = ax^n$

then graph of $\log y$ vs $\log x$ will give a straight line

where vertical intercept = $\log a$

gradient = n

birthday = year

	Relative distance from the sun (x)	revolution time (y)	$\log x$	$\log y$	
Mercury	0.39	0.24	-0.409	-0.620	
Venus	0.72	0.62	-0.143	-0.209	
Earth	1.00	1.00	0	0	
Mars	1.52	1.88	0.182	0.274	
Saturn	9.54	29.5	0.980	1.470	

3 sf
assume that
 $y = ax^n$

now

⇒ Proof: $y = ax^n$

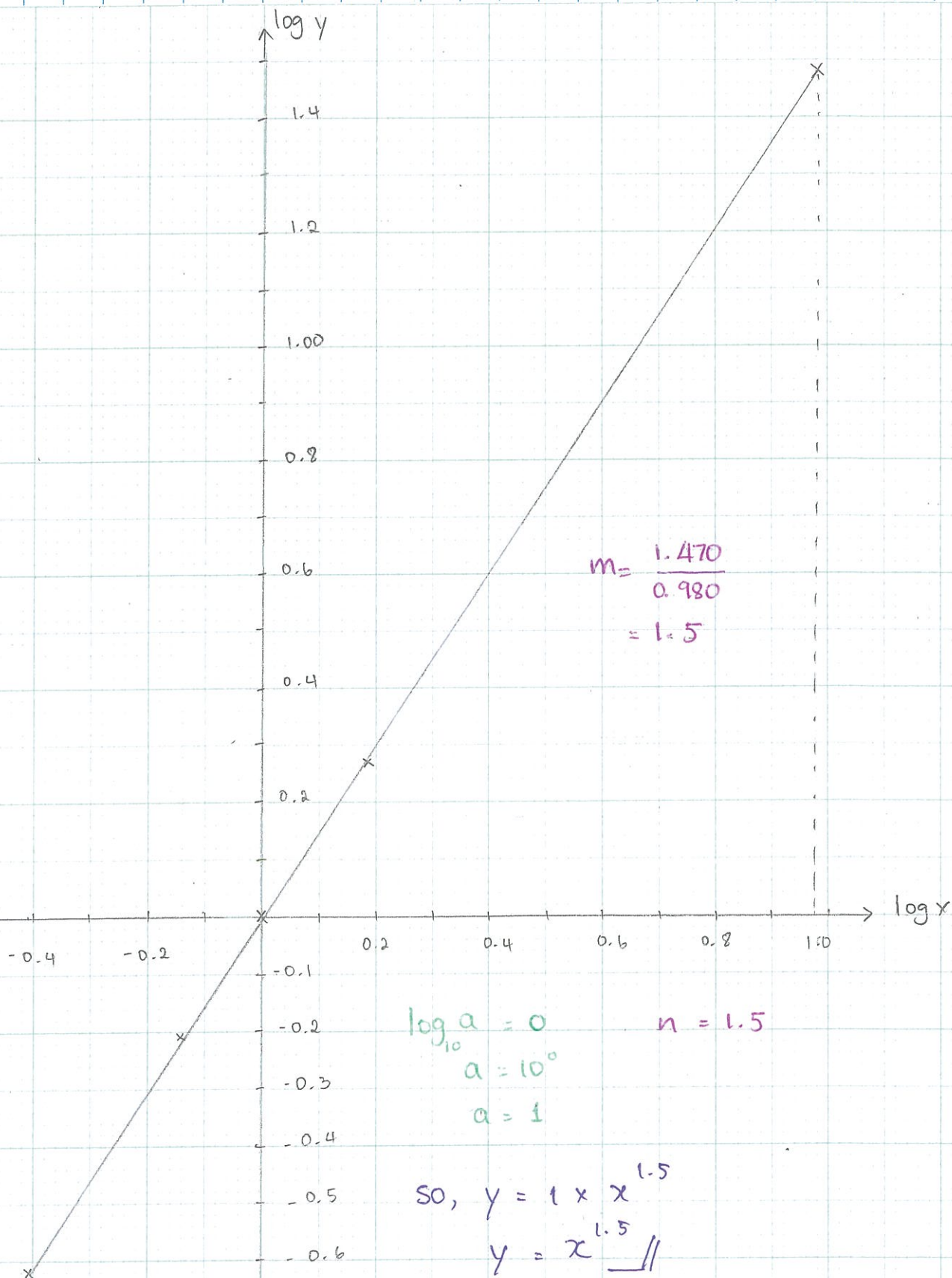
$$\log(y) = \log(ax^n)$$

$$= \log(a) + \log(x^n)$$

$$= \log(a) + n \log(x)$$

$$\log(y) = \underset{m}{n} \underset{x}{\log(x)} + \underset{c}{[\log(a)]}$$

$\log_{10}$ (base 10)



Cambridge style Q's on Logs and exponents

1. Equations with several log terms

Technique for solving

Try and rearrange into either

$$\log_b \square = \text{number}$$

then $\square = b^{\text{number}}$

or

$$\log_b \overset{f(x)}{\square} = \log_b \overset{g(x)}{\triangle}$$

then $\square = \triangle$

Eg. solve $\ln(3-x) = 4 + \ln x$

X $\ln 3 - \ln x$ $\ln(3-x) - \ln(x) = 4$

$$\log_e \square = 4$$

$$\square = e^4$$

$$\ln\left(\frac{3-x}{x}\right) = 4$$

$$\frac{3-x}{x} = e^4$$

$$3-x = e^4 x$$

$$e^4 x + x = 3$$

$$x(e^4 + 1) = 3$$

$$x = \frac{3}{e^4 + 1}$$

(exact value)

2. Equations with several exponential terms

e.g. $3^{2x} + 3^x = 12$.

Technique:

write all the exponential terms in terms of one of the exponentials. Then use another variable for this exponential.

Note: $3^{2x} = (3^x)^2$

So, $3^{2x} + 3^x = 12$

$(3^x)^2 + 3^x = 12$

Let $u = 3^x$

$u^2 + u = 12$

cannot find

$u^2 + u - 12 = 0$

log with

$(u+4)(u-3) = 0$

(-)

$u = -4$ or $u = 3$



so $3^x = -4$ or $3^x = 3$

no solution

$x = 1$

So, $x = 1$.

use logarithms to solve the equation

$$5^{2x-1} = 2(3^x)$$

giving your answer correct to 3 significant figures.

log both side

$$\log(5^{2x-1}) = \log(2(3^x))$$

$$(2x-1) \log 5 = \log 2 + \log 3^x$$

$$(2 \log 5)x - \log 5 = \log 2 + x \log 3$$

$$(\log 5^2)x - (\log 3)x = \log 2 + \log 5$$

$$x(\log 25 - \log 3) = \log(10)^{2 \times 5}$$

$$x = \frac{\log(10)}{\log(25) - \log(3)} = 1.08599$$

$$x = 1.09$$